

School of Computer Science
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BEI PENG

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RESEARCH INTERESTS

Deep Reinforcement Learning, Interactive Machine Learning, and Multi-Agent Systems

EDUCATION

- **Doctor of Philosophy**, Computer Science. Aug. 2013 – Jul. 2018
Washington State University, Pullman, WA, United States
Advisor: *Matthew E. Taylor* (now at the University of Alberta)
Dissertation: Learning from Human Teachers: Supporting How People Want to Teach in Interactive Machine Learning
- **Bachelor of Science**, Computer Science. Sep. 2008 – Jun. 2012
Huazhong University of Science & Technology, Wuhan, Hubei, China

ACADEMIC EMPLOYMENT

- **Lecturer (Assistant Professor)** in Artificial Intelligence and Applications of AI Aug. 2025 – Present
University of Sheffield, Sheffield, United Kingdom
- **Lecturer (Assistant Professor)** in Artificial Intelligence Sep. 2021 – Jul. 2025
University of Liverpool, Liverpool, United Kingdom
- **Non-Stipendiary Lecturer** in Computer Science Nov. 2019 – Aug. 2021
University of Oxford, Oxford, United Kingdom
- **Postdoctoral Researcher** Jan. 2019 – Aug. 2021
University of Oxford, Oxford, United Kingdom
Research on deep reinforcement learning at the Whiteson Research Lab with *Prof. Shimon Whiteson*
- **Graduate Research Assistant** Jan. 2014 – Feb. 2018
Washington State University, Pullman, WA, United States
Research on interactive machine learning at the Intelligent Robot Learning Lab with *Prof. Matthew E. Taylor*

TEACHING EXPERIENCE

- **University of Sheffield**
 - COM2109: Automata, Computation, and Complexity, *Module Coordinator*, Fall 2025
 - COM3124: Bioinspired Computing, *Module Coordinator*, Fall 2025
- **University of Liverpool**
 - COMP310: Multi-Agent Systems, *Module Coordinator*, Spring 2022, 2023, 2024, 2025
 - Awarded the Postgraduate Certificate Academic Practice (PGCAP) with *Distinction*
 - Received the *Fellow of the Higher Education Academy (FHEA) Certificate* on 3 January 2024
- **University of Oxford**
 - Artificial Intelligence, *Tutor*, Spring 2020, Spring 2021

- Machine Learning, *Tutor*, Fall 2019, Fall 2020
- Reinforcement Learning, *Teaching Assistant*, Fall 2019, Fall 2020

- **Washington State University**

- Reinforcement Learning, *Teaching Assistant*, Spring 2015
- Introduction to Computer Architecture, *Teaching Assistant*, Fall 2013

SUPERVISION EXPERIENCE

- **University of Sheffield** Aug. 2025 – Present
 - I am currently co-supervising 7 PhD students from the University of Liverpool.
- **University of Liverpool** Sep. 2021 – Jul. 2025
 - I have supervised 15 master’s projects from the academic years 2021-22 to 2023-24.
 - I have supervised 27 undergraduate final year projects from the academic years 2021-22 to 2024-25.
 - I have hosted 3 PhD visiting students, two from XJTLU in China, one from Sumy State University, which is funded by the UK-Ukraine R&I Twinning Grant Scheme.
 - I have supervised an undergraduate student internship funded by EPSRC 2024 Vacation Internship Scheme.
- **University of Oxford** Jan. 2019 – Aug. 2021
 - I have (co-)supervised 6 PhD students and 1 master’s student within/outside Oxford:
 - Tabish Rashid, Christian Schroeder de Witt, Tarun Gupta, Jacob Beck (all from University of Oxford)
 - Shariq Iqbal (University of Southern California)
 - Ling Pan, Tonghan Wang (both from Tsinghua University)
 - I have supervised 4 undergrads in Oxford: Bozhida Vasilev, Kaloyan Aleksiev, Benjamin Slater, Leo Feng.

INDUSTRY EXPERIENCE

- **Microsoft Research**, Redmond, WA, United States Oct. 2018 – Dec. 2018
Research Intern
 Focused on developing hierarchical deep reinforcement learning algorithms to learn interactive fiction games.
- **Borealis AI**, Edmonton, Alberta, Canada Mar. 2018 – Jun. 2018
Research Intern
 Focused on developing algorithms to learn sequential decision-making tasks from online evaluative human feedback.
- **Tencent AI**, Seattle, WA, United States Aug. 2017 – Nov. 2017
Research Intern
 Focused on training the agent to play MOBA game King of Glory using deep supervised learning and RL algorithms.
- **Tencent**, Wuhan, Hubei, China Jun. 2012 – May 2013
Front-End Web Developer
 Implemented web extensions and web games in mobile platform by JavaScript.

PUBLICATIONS

Journal Articles

Flora Charbonnier, **Bei Peng**, Julie Vienne, Elena Stai, Thomas Moisten, and Malcolm McCulloch. Centralised Rehearsal of Decentralised Cooperation: Multi-Agent Reinforcement Learning for the Scalable Coordination of Residential Energy Flexibility. *Applied Energy*, 2025.

Hui Jiao, **Bei Peng**, Lu Zong, Xiaojun Zhang, and Xinwei Li. Gradable ChatGPT Translation Evaluation. *Journal Procesamiento del Lenguaje Natural*, 2024.

Xiaowei Huang, **Bei Peng**, and Xingyu Zhao. Dependable Learning-Enabled Multiagent Systems. *AI Communications*, 2022.

Sanmit Narvekar, **Bei Peng**, Matteo Leonetti, Jivko Sinapov, Matthew E. Taylor, and Peter Stone. Curriculum Learning for Reinforcement Learning Domains: A Framework and Survey. *Journal of Machine Learning Research (JMLR)*, 2020.

Bei Peng, James MacGlashan, Robert Loftin, Michael L. Littman, David L. Roberts, and Matthew E. Taylor. Curriculum Design for Machine Learners in Sequential Decision Tasks. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2018.

Robert Loftin, **Bei Peng**, James MacGlashan, Michael L. Littman, Matthew E. Taylor, Jeff Huang, and David L. Roberts. Learning Behaviors via Human-Delivered Discrete Feedback: Modeling Implicit Feedback Strategies to Speed Up Learning. *Journal of Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 2016.

Conference Papers

Tianhui Zhang, **Bei Peng**, and Danushka Bollegala. Evaluating the Evaluation of Diversity in Commonsense Generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

Zhenglin Huang, Jinwei Hu, Yiwei He, Xiangtai Li, Xiaowei Huang, **Bei Peng**, Xingyu Zhao, Baoyuan Wu, and Guangliang Cheng. SIDA: Social Media Image Deepfake Detection, Localization and Explanation with Large Multimodal Model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

Tianhui Zhang, **Bei Peng**, and Danushka Bollegala. Improving Diversity of Commonsense Generation by Large Language Models via In-Context Learning. In *Proceedings of the Findings of Empirical Methods in Natural Language Processing (EMNLP)*, 2024.

Oliver Dippel, Alexei Lisitsa, and **Bei Peng**. Contextual Transformers for Goal-Oriented Reinforcement Learning. In *Proceedings of the SGAI International Conference on Artificial Intelligence*, 2024.

Bettina Hanlon, Ángel F. García-Fernández, and **Bei Peng**. A Comparison Between Kalman-MLE and KalmanNet for State Estimation with Unknown Noise Parameters. In *Proceedings of the IEEE International Conference on Multisensor Fusion and Integration (MFI)*, 2024.

Gabriella Pizzuto, Hetong Wang, Hatem Fakhruddin, **Bei Peng**, Kevin Sebastian luck, and Andrew Ian Cooper. Accelerating Laboratory Automation Through Robot Skill Learning for Sample Scraping. In *Proceedings of the IEEE 20th International Conference on Automation Science and Engineering (CASE)*, 2024. **Best HealthCare Automation Paper Finalist**.

Tianhui Zhang, Danushka Bollegala, and **Bei Peng**. Learning to Predict Concept Ordering for Common Sense Generation. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AACL)*, 2023.

Oliver Dippel, Alexei Lisitsa, and **Bei Peng**. Deep Reinforcement Learning for Continuous Control of Material Thickness. In *Proceedings of the SGAI International Conference on Artificial Intelligence*, 2023.

Bei Peng*, Tabish Rashid*, Christian A. Schroeder de Witt*, Pierre-Alexandre Kamienny, Philip H. S. Torr, Wendelin Böhmer, and Shimon Whiteson. FACMAC: Factored Multi-Agent Centralised Policy Gradients. *In Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS)*, 2021.

Ling Pan, Tabish Rashid, **Bei Peng**, Longbo Huang, Shimon Whiteson. Regularized Softmax Deep Multi-Agent Q-Learning. *In Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS)*, 2021.

Shariq Iqbal, Christian A. Schroeder de Witt, **Bei Peng**, Wendelin Böhmer, Shimon Whiteson, and Fei Sha. Randomized Entity-wise Factorization for Multi-Agent Reinforcement Learning. *In Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.

Tarun Gupta, Anuj Mahajan, **Bei Peng**, Wendelin Böhmer, and Shimon Whiteson. UneVEN: Universal Value Exploration for Multi-Agent Reinforcement Learning. *In Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.

Tonghan Wang, Tarun Gupta, Anuj Mahajan, **Bei Peng**, Shimon Whiteson, and Chongjie Zhang. RODE: Learning Roles to Decompose Multi-Agent Tasks. *In Proceedings of the 9th International Conference on Learning Representations (ICLR)*, 2021.

Tabish Rashid, Gregory Farquhar, **Bei Peng**, and Shimon Whiteson. Weighted QMIX: Expanding Monotonic Value Function Factorisation. *In Proceedings of the 34th Conference on Neural Information Processing Systems (NeurIPS)*, 2020.

Tabish Rashid, **Bei Peng**, Wendelin Böhmer, Shimon Whiteson. Optimistic Exploration even with a Pessimistic Initialisation. *In Proceedings of the 8th International Conference on Learning Representations (ICLR)*, 2020.

James MacGlashan, Mark Ho, Robert Loftin, **Bei Peng**, Guan Wang, David L. Roberts, Matthew E. Taylor, and Michael L. Littman. Interactive Learning from Policy-Dependent Human Feedback. *In Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

Bei Peng, James MacGlashan, Robert Loftin, Michael L. Littman, David L. Roberts, and Matthew E. Taylor. Curriculum Design for Machine Learners in Sequential Decision Tasks. *In Proceedings of the 16th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2017.

Bei Peng, James MacGlashan, Robert Loftin, Michael L. Littman, David L. Roberts, and Matthew E. Taylor. A Need for Speed: Adapting Agent Action Speed to Improve Task Learning from Non-Expert Humans. *In Proceedings of the 15th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2016.

Gabriel V. de la Cruz Jr., **Bei Peng**, Walter S. Lasecki, and Matthew E. Taylor. Towards Integrating Real Time Crowd Advice with Reinforcement Learning. *In proceedings of the 20th ACM Conference on Intelligent User Interfaces (IUI)*, 2015.

Robert Loftin, **Bei Peng**, James MacGlashan, Michael L. Littman, Matthew E. Taylor, David Roberts, and Jeff Huang. Learning Something from Nothing: Leveraging Implicit Human Feedback Strategies. *In IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, 2014.

Robert Loftin, James MacGlashan, **Bei Peng**, Michael L. Littman, Matthew E. Taylor, Jeff Huang, and David L. Roberts. A Strategy-Aware Technique for Learning Behaviors from Discrete Human Feedback. *In Proceedings of the 28th AAAI Conference on Artificial Intelligence (AAAI)*, 2014.

Workshop and Symposium Papers

Jingjing Feng, Lucheng Wang, Alona Tenytska, and **Bei Peng**. PRIDE: Preference-Based Reinforcement Learning Using Diffusion Model. *In Proceedings of the Adaptive and Learning Agents Workshop (at AAMAS)*, 2025.

Lin Shi and **Bei Peng**. Curriculum Learning for Relative Overgeneralization. *In Proceedings of the Adaptive and Learning Agents Workshop (at AAMAS)*, 2023.

Bozhidar Vasilev, Tarun Gupta, **Bei Peng**, and Shimon Whiteson. Semi-On-Policy Training for Sample Efficient Multi-Agent Policy Gradients. *In Proceedings of the Adaptive and Learning Agents Workshop (at AAMAS)*, 2021.

Leo Feng, Luisa Zintgraf, **Bei Peng**, and Shimon Whiteson. VIABLE: Fast Adaptation via Backpropagating Learned Loss. *In Proceedings of the 3rd Workshop on Meta-Learning (at NeurIPS)*, 2019.

Tabish Rashid, **Bei Peng**, Wendelin Bohmer, and Shimon Whiteson. Optimistic Exploration with Pessimistic Initialization. *In Proceedings of the Exploration in Reinforcement Learning Workshop (at ICML)*, 2019.

Bei Peng, James MacGlashan, Robert Loftin, Michael L. Littman, David L. Roberts, and Matthew E. Taylor. Curriculum Design for Machine Learners in Sequential Decision Tasks. *In Proceedings of the Adaptive Learning Agents Workshop (at AAMAS)*, 2017.

Robert Loftin, James MacGlashan, **Bei Peng**, Matthew E. Taylor, Michael L. Littman, and David L. Roberts. Towards Behavior-Aware Model Learning from Human-Generated Trajectories. *In AAAI Fall Symposium on Artificial Intelligence for Human-Robot Interaction*, 2016.

James MacGlashan, Michael L. Littman, David L. Roberts, Robert Loftin, **Bei Peng**, and Matthew E. Taylor. Convergent Actor Critic by Humans. *In Workshop on Human-Robot Collaboration: Towards Co-Adaptive Learning Through Semi-Autonomy and Shared Control (at IROS)*, 2016.

Bei Peng, James MacGlashan, Robert Loftin, Michael L. Littman, David L. Roberts, and Matthew E. Taylor. An Empirical Study of Non-Expert Curriculum Design for Machine Learners. *In Proceedings of the Interactive Machine Learning Workshop (at IJCAI)*, 2016.

Mitchell Scott, **Bei Peng**, Madeline Chili, Tanay Nigam, Francis Pascual, Cynthia Matuszek, and Matthew E. Taylor. On the Ability to Provide Demonstrations on a UAS: Observing 90 Untrained Participants Abusing a Flying Robot. *In Proceedings of the AAAI Fall Symposium on Artificial Intelligence and Human Robot Interaction AI-HRI*, 2015.

Bei Peng, Robert Loftin, James MacGlashan, Michael L. Littman, Matthew E. Taylor, and David L. Roberts. Language and Policy Learning from Human-delivered Feedback. *In proceedings of the Machine Learning for Social Robotics Workshop (at ICRA)*, 2015.

Gabriel V. de la Cruz Jr., **Bei Peng**, Walter S. Lasecki, and Matthew E. Taylor. Generating Real-Time Crowd Advice to Improve Reinforcement Learning Agents. *In Proceedings of the Learning for General Competency in Video Games workshop (at AAAI)*, 2015.

James Macglashan, Michael L. Littman, Robert Loftin, **Bei Peng**, David L. Roberts, and Matthew E. Taylor. Training an Agent to Ground Commands with Reward and Punishment. *In Proceedings of the Machine Learning for Interactive Systems workshop (at AAAI)*, 2014.

SELECTED TALKS

- Advancing AI through Collaboration: From Multi-Agent to Human-in-the-loop Reinforcement Learning
Invited Talk at the Winter School on Robotics & AI for Materials Discovery at Liverpool, December 2024.
- Relative Overgeneralisation in Cooperative Multi-Agent Reinforcement Learning
Invited Seminar Talk at the University of Sheffield, Feb 2024.
- Overcoming Relative Overgeneralisation for Cooperative Multi-Agent Reinforcement Learning
Invited Talk at the Game Theory and Machine Learning Workshop at London School of Economics, Oct 2023.

- Factored Multi-Agent Centralised Policy Gradients
Guest Talk at the Centre for Doctoral Training in Distributed Algorithms at Liverpool, June 2023.
- Introduction to Reinforcement Learning
Tutorial at the Centre for Doctoral Training in Distributed Algorithms at Liverpool, November 2022.
- Introduction to Multi-Agent Reinforcement Learning
Invited Lecture at Canada CIFAR 2022 Deep Learning+Reinforcement Learning Summer School, July 2022.
- Cooperative Multi-Agent Reinforcement Learning
Invited Keynote Talk at the Adaptive Learning Agents (ALA) Workshop at AAMAS, May 2022.
- Cooperative Deep Multi-Agent Reinforcement Learning
Invited Talk at the Centre for Mathematical Imaging Techniques Seminar, University of Liverpool, March 2022.
- Learning from Evaluative Human Feedback
Invited Keynote Talk at the Transparent Agency and Learning Workshop, September 2021.
- FACMAC: Factored Multi-Agent Centralised Policy Gradients
Paper presentation in 35th Conference on Neural Information Processing Systems (NeurIPS), December 2021.
- Analytic Multi-Agent Actor-Critic Algorithms
Talk at the Whiteson Research Lab, University of Oxford, April 2020.
- Learning Behaviors via Human-Delivered Discrete Feedback
Talk at the Whiteson Research Lab, University of Oxford, February 2019.
- Learning from Human Teachers: Supporting How People Want to Teach in Interactive Machine Learning
Invited Talk at Microsoft Research, Redmond, WA, United States, July 2018.
- Curriculum Design for Machine Learners in Sequential Decision Tasks
Paper presentation in the Conference on Autonomous Agents and Multi-agent Systems (AAMAS), May 2017.
- A Need for Speed: Adapting Agent Action Speed to Improve Task Learning from Non-Expert Humans
Paper presentation in the Conference on Autonomous Agents and Multi-agent Systems (AAMAS), May 2016.

AWARDS AND HONORS

- Grace Hopper Celebration Faculty Scholarship, 2023
- SU EECS Scholarship for Grace Hopper Celebration Conference, 2017
- AAMAS NSF Scholarship, 2016
- RSJ/KROS Distinguished Interdisciplinary Research Award Finalist for our paper “Learning something from nothing: Leveraging implicit human feedback strategies” at RO-MAN 2014
- Travel Award:
 - Conference on Autonomous Agents and Multi-agent Systems (AAMAS) 2016, 2017
 - International Joint Conferences on Artificial Intelligence (IJCAI) 2016
 - Grad Cohort Workshop for Women 2014, 2015
 - AAAI Conference on Human Computation and Crowdsourcing (HCOMP) 2014
- National Encouragement Scholarship (1%), Huazhong University of Science and Technology, China, 2011
- Model Student of Academic Records (1%), Huazhong University of Science and Technology, China, 2010
- Individual Scholarship (5%), Huazhong University of Science and Technology, China, 2009

GRANTS AND FUNDING

The Royal Society International Exchanges 2024 Cost Share (NSFC) Scheme.

01/2025-12/2026

Role: PI, £12,000.

“Graph-Based Deep Reinforcement Learning for Resilience-Constrained Dynamic Dispatch of Smart Grids.”

ACADEMIC SERVICE

- Co-Chair for the Workshops/Tutorials at the International Conference on Distributed Artificial Intelligence (DAI) 2025.
- Co-Chair for the Competition Track of International Joint Conference on Artificial Intelligence (IJCAI) 2024.
- IPAP (Independent Progress Assessment Panel) Member for 17 PhD students at the University of Liverpool from 2021-2025.
- Search Committee Member for a new Dean in the School of Electrical Engineering, Electronics and Computer Science at the University of Liverpool, 2024.
- Senior Program Committee Member for International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2024, 2025
- Reviewer for the Canada CIFAR AI Chairs Program 2022, 2023, 2024, 2025.
- Panellist in the RL-CONFROM Workshop at the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023
- Search Committee Member for academic posts (T&R) in Department of Computer Science, University of Liverpool, 2022, 2023.
- PhD Thesis External Examiner for Alex Beeson, University of Warwick, May 2025
- PhD Thesis Internal Examiner for Tianle Zhang, University of Liverpool, Feb 2025
- PhD Thesis External Examiner for Taku Yamagata, University of Bristol, Nov 2024
- PhD Thesis Internal Examiner for Dumitru Mirauta, University of Liverpool, Sep 2024
- PhD Thesis External Examiner for Simon Vanneste, University of Antwerp, Jul 2024
- PhD Thesis Internal Examiner for Emmanouil Pitsikalis, University of Liverpool, May 2024
- PhD Thesis Internal Examiner for Andrew Roxburgh, University of Liverpool, Nov 2023
- PhD Thesis Internal Examiner for Samantha Durdy, University of Liverpool, Oct 2023
- PhD Thesis External Examiner for Canmanie Ponnambalam, Delft University of Technology, Jun 2023
- PhD Thesis Internal Examiner for James Butterworth, University of Liverpool, Feb 2023
- External Panel Member for Postdoc Interviews at the Chemistry Department, University of Liverpool, 2022
- Panel Member for PhD Interviews at the Whiteson Research Lab, University of Oxford, 2020, 2021
- Panel Member for Undergraduate Admission at the St Catherine’s College, University of Oxford, 2019, 2020
- Co-organizer (with Patrick MacAlpine, Patrick Mannion, and Roxana Radulescu), Adaptive Learning Agents (ALA) Workshop at AAMAS 2019
- Co-organizer (with Anna Harutyunyan, Patrick Mannion, and Kaushik Subramanian), Adaptive Learning Agents (ALA) Workshop at AAMAS 2018

- Conference and Journal Reviewing:
 - International Conference on Learning Representations (ICLR) 2021, 2022, 2023, 2024
 - International Conference on Intelligent Robots and Systems (IROS) 2023, 2024
 - Journal of Autonomous Agents and Multi-Agent Systems (JAAMAS) 2024
 - IEEE International Conference on Automation Science and Engineering (CASE) 2024
 - Learning for Dynamics and Control (L4DC) Conference 2024
 - Doctoral Consortium at AAAI Conference on Artificial Intelligence (AAAI) 2024
 - Coordination and Cooperation in Multi-Agent Reinforcement Learning Workshop at RLC 2024
 - Adaptive Learning Agents Workshop (ALA) at AAMAS 2017, 2018, 2019, 2021, 2023, 2024
 - IEEE Transactions on Pattern Analysis and Machine Learning, 2022, 2023
 - Neuro-Symbolic AI for Agent and Multi-Agent Systems Workshop at AAMAS 2023
 - Conference on Neural Information Processing Systems (NeurIPS) 2020, 2021
 - Journal of Machine Learning Research (JMLR) 2021
 - International Journal of Computer Vision (IJCV) 2021
 - Journal of Artificial Intelligence Research (JAIR) 2020
 - AAAI Conference on Artificial Intelligence (AAAI) 2020
 - International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2019
 - Scaling-Up Reinforcement Learning Workshop (SURL) at ECML PKDD, 2017, 2019
 - IEEE Geoscience Remote Sensing Letters, 2017
 - Workshop on the Future of Interactive Learning Machines (FILM) at NeurIPS, 2016